

Question	Marking Guidance	Mark	Comments
07.1	1. Mass of DNA equals 1 (indicates) (early) interphase OR No cells with mass of DNA less than 1 (indicates) mitosis only type of division occurring; 2. Mass of DNA equals 2 (indicates) mitosis OR Mass of DNA equals 2 (indicates) each chromosome consists of 2 chromatids; 3. Mass of DNA between 1 and 2 (indicates) some of the DNA has been replicated;	3 (3 x AO3)	1. Accept G1 phase Accept no meiosis 2. Accept G2 / any named phase of mitosis 3. Accept S phase

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07.2	1. Fewer cells with mass of DNA = 1 OR More cells with mass of DNA between 1 and 2 OR More cells with mass of DNA 2; 2. Cell division is faster/uncontrolled OR More mitosis/cell division/cell replication;	2 (2 x AO3)	'Mass of DNA is higher' is insufficient 2. Ignore 'growth' 2. Accept 'uncontrolled mitosis'

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07.3	1. (Chromosomes) condense; 2. (Chromosomes) line up on equator/centre of cell OR (Chromosomes) attached to spindle;	2 (2 x AO1)	1. Accept 'shorten' or 'thicken'

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07.4	Correct answer of 20 = 2 marks ;; 1 mark for evidence of $2 \times 10^{-8} \text{ (m s}^{-1}\text{)}$	2 (2 x AO2)	